Meta-analysis

Master of Statistics
KU Leuven

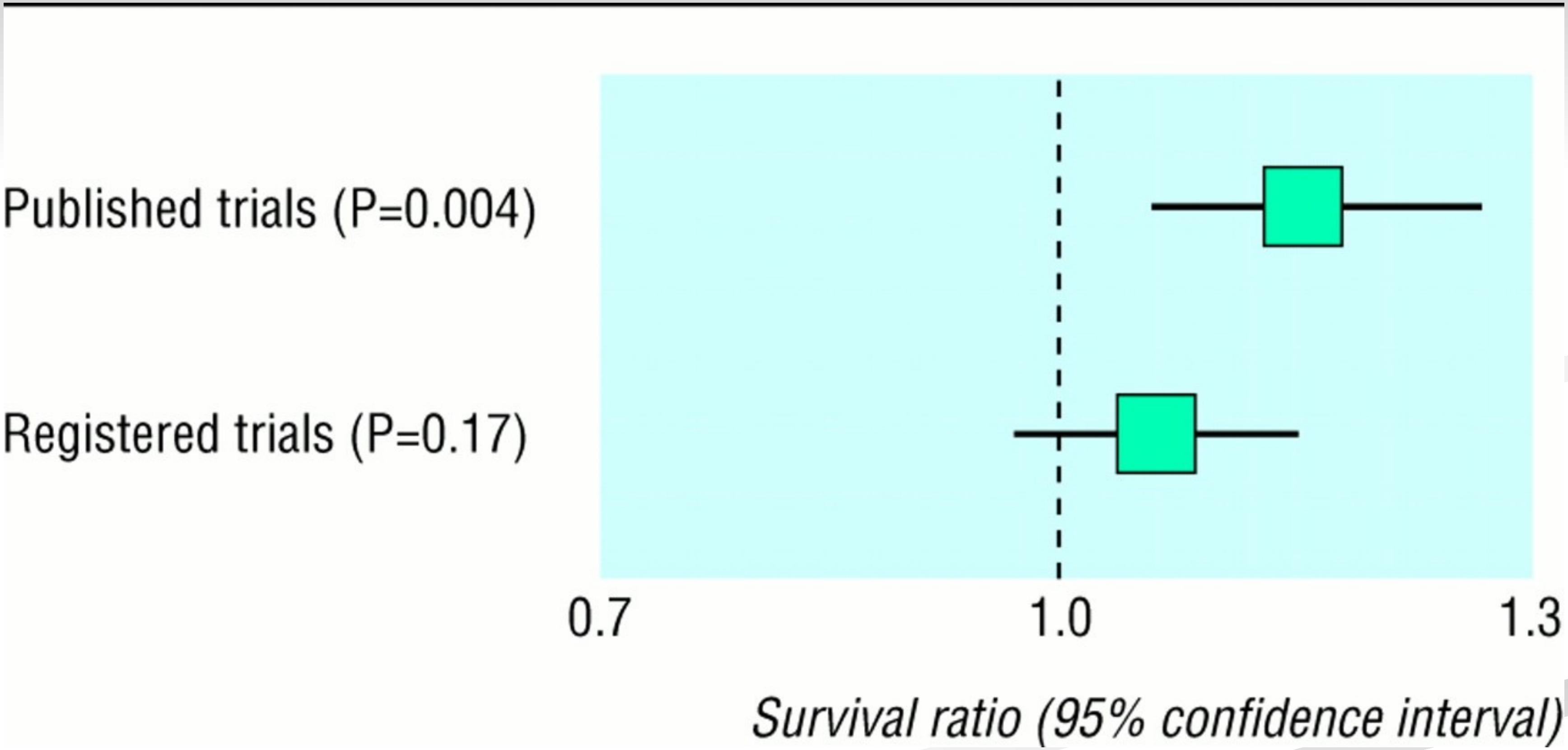
Avoiding bias

Be aware of bias!

- Outcome reporting bias
- Publication bias
- Citation bias
- Multiple publication bias
- Bias in keywords
- Bias in sources that are used
- Bias in inclusion criteria (e.g., study design, …)
- Geographical/language bias
- ….

Example 1: advanced ovarian cancer:
monotherapy alkylating agent vs. combination chemotherapy?

International Cancer Research Data Bank



(Egger, M. D., & Smith, G. (1998). Meta-analysis. Bias in location and selection of studies. *British Medical Journal, 316*, 61-66. http://www.bmj.com/cgi/content/full/316/7124/61).

Example 2:
510 large trials presented at conferences of the American Society of Clinical Oncology (ASCO)

Proportion of publication within 5 years after conference:

- 81 % (of 233 trials) for significant results
- 68 % (of 287 trials) for nonsignificant results

(Kryzanowska, M. K., Pintilie, M., & Tennock, I. F. (2003). Factors associated with failure to publish large randomized trials presented at an oncology meeting. *Journal of the American Medical Association, 290*, 495-501).

Example 3:
About 95 % of tested null hypotheses in a random sample of papers in psychology & psychiatry were rejected (Fanelli, 2012) ↔ estimated average power in psychological domain is .35 (Bakker, Van Dijk, & Wicherts, 2012) or .47 (Cohen, 1990)

Example 4:
‘The Reproducibility Project’ (Open Science Collaboration, 2015). 97 out of 100 large studies in major journals found significant results; only 35 replications found significant results.

Example 5:
Many reported statistical results just reach statistical significance.
Distribution for 1,000 randomly drawn psychological studies in 2007:

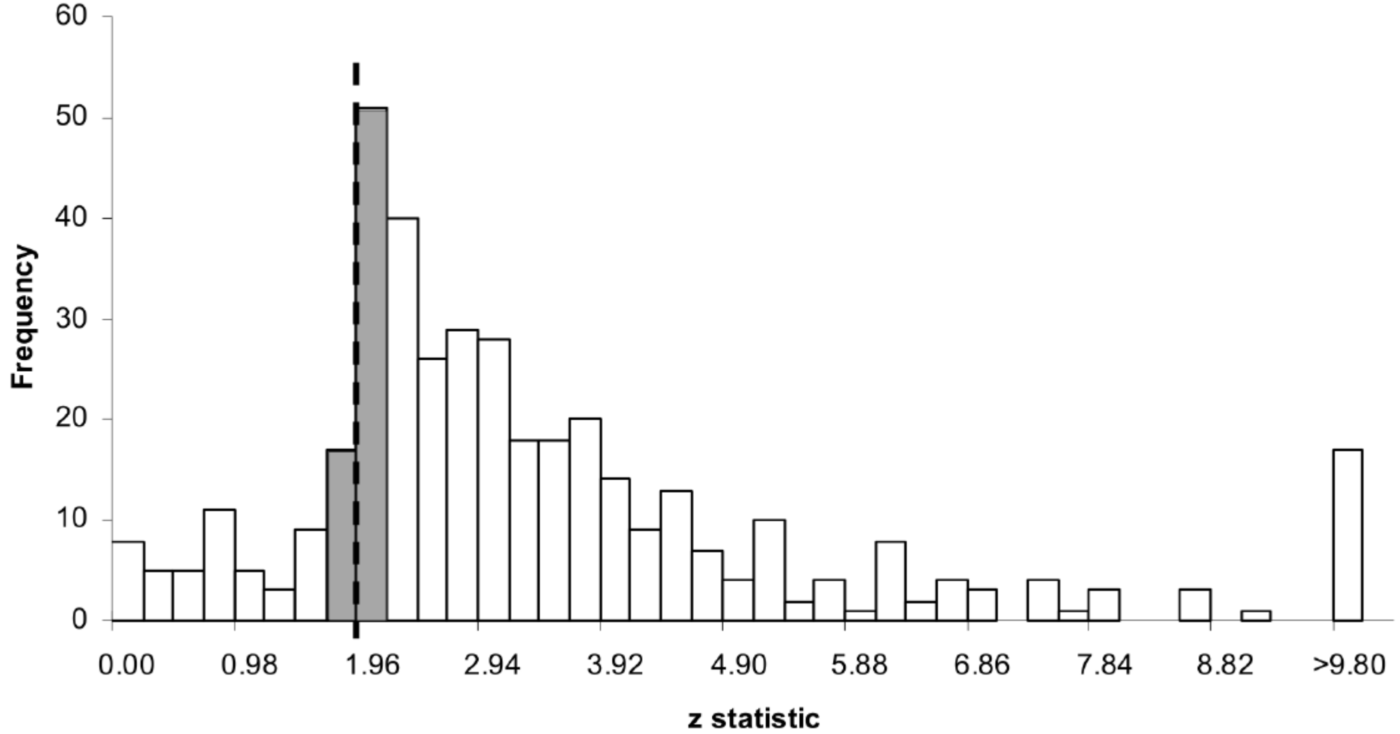


Figure 6. Distribution of z-transformed p values. Note: Dashed line specifies the critical z-statistic (1.96) associated with p = .05 significance level for two-tailed tests. Width of intervals (0.245 i.e. a multiple of 1.96) correspond to a 12.5% caliper.

(Kühberger, A., Fritz, A., & Scherndl, T. (2014). Publication bias in psychology: A diagnosis based on the correlation between effect size and sample size. *PLoS ONE, 9*(9), e105825.

'A pooled weighted average of 1.97% (N = 7, 95%CI: 0.86–4.45) of scientists admitted to have ***fabricated, falsified or modified data*** *or results at least once –a serious form of misconduct by any standard– and up to 33.7% admitted other questionable research practices.'*

Fanelli, D. (2009). How many scientists fabricate and falsify research? A systematic review and meta-analysis of survey data. *PLoS ONE, 4*(5): e5738.

Or (somewhat) less serious: ***p*-hacking** (using researchers' df to obtain statistically significant results).

O'Boyle, E.H., Gonzalez-Mule, E., & Banks, G.C. (2017). The Chrysalis effect: How ugly initial results metamorphosize into beautiful articles. *Journal of Management, 43*(2), 376-399.

Letter from editor scientic journal October 30, 2014

Dear Mrs. XXXXX,

I have completed an initial reading of your manuscript, which has been submitted for publication to *XXXXXXXXXXX*. Before sending manuscripts out for full review, I conduct a preliminary screening to determine both the appropriateness of manuscript in light of the journal's mission as well as its likely competitiveness for publication. This can save authors and reviewers valuable time.

On the basis of this evaluation, I have decided not to send your manuscript out for full review. My reading of your manuscript indicates that while the content fits the scope of the journal, it is very unlikely to be acceptable for publication for the several reasons. Perhaps the greatest limitation of your manuscript is the weakness of the reported findings. Although a small number of the gene-by-environment interactions are statistically significant, particularly after controlling for the number of statistical tests performed, these effects are very small in magnitude. For example, among those interactions that are statistically significant (i.e., discounting those that do not meet conventional levels of significance) none led to an increase in the R2 greater than .01. That is, none of these effects account for more than 1% of the variance in the outcome measure. These effects, while statistically significant, could be interpreted as relatively trivial by readers.

Letter from editor scientic journal January 30, 2018

Dear Ms. *XXXX*,

Thank you for revising and resubmitting your manuscript for publication in *XXX*. .... After carefully reading your manuscript, I have decided it does not warrant further consideration for publication in *XXX*. ...

Like the reviewers, I thought you did a very thorough job addressing the various concerns raised about your original manuscript. I believe you addressed all of them, except for the concerns regarding measuring engagement once-a-year, and the question regarding whether your findings would make a sufficient contribution to the literature to warrant publication in CEP. I think the first concern, which cannot be addressed through revision, aggravated the second concern. If your findings had been particularly large or striking, then having only measured engagement once a year might not have been a concern. But, your relatively modest findings call into question whether there were significant changes that occurred between measurement points that were missed. ...I am confident this manuscript will be published, and I think you did a fine job overall, but the significance of the manuscript's findings and conclusions do not rise to the level necessary to be published in *XXX*.

Discussion:

“God loves the .06 (nearly) as much as the .05”
(Rosnow R, & Rosenthal R. (1989). Statistical procedures and the justification of knowledge in psychological science. *American Psychologist, 44,* 1276–1284)

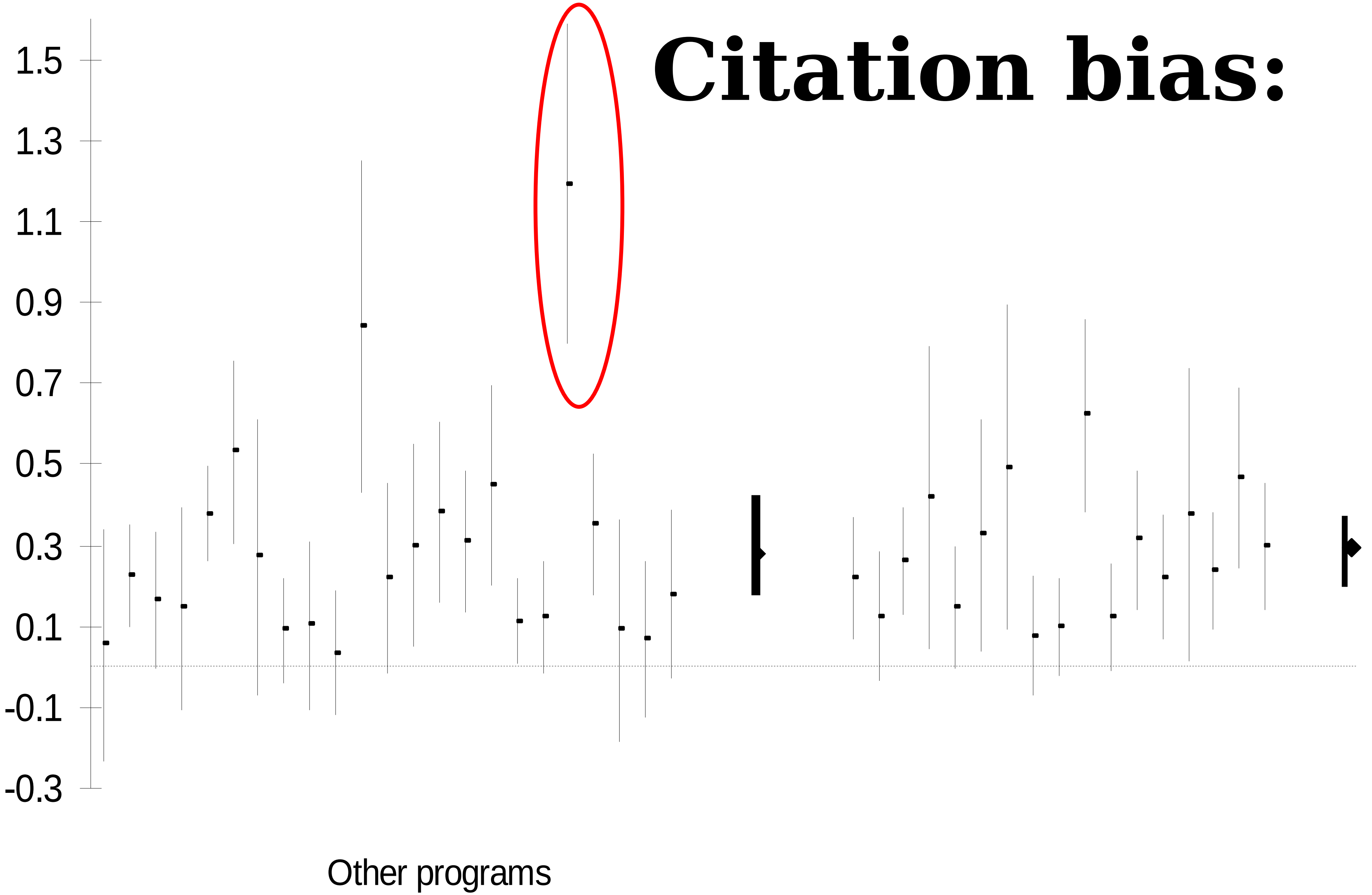
vs.

nonsignificant results may be uninteresting:

a) **Underpowered studies**
b) **Implausible hypotheses**
c) **Invalid operationalization of the variables**
d) **…**

(see Dick et al. (2015). Candidate Gene-Environment interaction research: Reflections and recommendations. *Perspectives on Psychological Science, 10,* 37-59)

Citation bias:



Geeraert, L., Van den Noortgate, W., Grietens, H., & Onghena, P. (2004). The effects of early prevent programs for families with young children at risk for physical child abuse and neglect. A meta-analy *Chi Maltreatment, 9, 277-291.*

Avoid bias

- Techniques to detect bias or correct for bias are not waterproof
- Try to find unpublished literature:
  - Make use of databases for ‘grey’ literature, such as:
    - European OpenGrey database (https://opengrey.eu/)
    - ProQuest Dissertations & Theses Global
  - Possibly contact research groups for asking non-published results
- Avoid bias in your own (primary and [illegible] analytic) research: report all relevan[illegible] regardless of their significance